Corporate Finance

Lecture 3 extras: Read the news, and look around you

Camilo Saldarriaga

Finance Department, ESCP Business School, 2026/2027

Extra 1

ENI

Eni: a big dividend payer in Milan



Palazzo Eni, Rome

- Italian oil and gas group, listed in Milan and New York. The state holds about 30%.
- Share price about **EUR 24**, market cap about **EUR 60 billion**.
- 2025: adjusted net profit **EUR 5.0 billion**, operating cash flow **EUR 12.5 billion**.
- Returned **EUR 5.0 billion** to shareholders in 2025: EUR 3.1 billion of dividends, EUR 1.9 billion of buybacks.
- Dividend of **EUR 1.05 per share**: a **yield of about 4.5%**.

Sources: Eni 2025 results, dividend policy; market data Sept. 2026.

Extra 2

Indices and dividends

Which indices include dividends?

Gross total return: dividends reinvested	Price only
DAX 40 (Germany)	S&P 500, Dow Jones, Nasdaq (US)
Ibovespa (Brazil)	FTSE 100 (UK)
	Nikkei 225 (Japan)
	CAC 40 (France)
	Hang Seng (Hong Kong)

- **Price index**: when a company pays a dividend, its share price drops, and the index does not add that cash back. It only measures price changes.
- **Total return index**: assumes every dividend is reinvested in the index, before tax, so it grows faster over time.
- Most price indices also have an official total return version, e.g. the S&P 500 TR or the CAC 40 GR.

Types of indeces

- **Price index** (price-only or capital index): follows share prices only. When a company pays a dividend its price drops and the index drops with it; the cash is never added back. The headline S&P 500, Dow Jones, FTSE 100, CAC 40 and Nikkei 225 are all price indices.
- **Gross total return index** (gross return or performance index): every dividend is reinvested in the index on the day it is paid, before any tax. Examples: the DAX 40 as quoted in the news, the S&P 500 TR, the CAC 40 GR.
- **Net total return index** (net return index): dividends are reinvested after the withholding tax a foreign investor would pay. The usual benchmark for international funds and ETFs. Examples: MSCI World Net, S&P 500 Net.

Extra 3

Variance and covariance, alternative expressions

Alternative expressions of the variance

- The variance is the average squared distance to the mean. For N observations $x_1, \ldots, x_N$ with mean $\bar{x}$:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

- The same number can be written without any distances, as the **mean of the squares minus the square of the mean**:

$$\mathbf{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} x_i^2 - \left(\frac{1}{N} \sum_{i=1}^{N} x_i \right)^2$$

- This is the **König-Huygens formula**. It is handy with a calculator: you add the values, add their squares, and never compute a single distance.

König-Huygens step by step

- Start from the definition and expand the square inside the sum:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i^2 - 2\,x_i\,\bar{x} + \bar{x}^2)$$

- Average each of the three pieces separately:

$$\sigma^2 = \underbrace{\frac{1}{N}\sum_{i=1}^{N} x_i^2}_{\text{mean of the squares}} - \underbrace{\frac{1}{N}\sum_{i=1}^{N} 2\,x_i\,\bar{x}}_{\text{middle term}} + \underbrace{\frac{1}{N}\sum_{i=1}^{N} \bar{x}^2}_{\text{last term}}$$

König-Huygens step by step (con't)

- Develop the middle and last terms:

$$\frac{1}{N}\sum_{i=1}^{N} 2\,x_i\,\bar{x} = 2\bar{x}\cdot\frac{1}{N}\sum_{i=1}^{N} x_i = 2\bar{x}\cdot\bar{x} = 2\bar{x}^2$$

$$\frac{1}{N}\sum_{i=1}^{N} \bar{x}^2 = \frac{1}{N}\cdot N\bar{x}^2 = \bar{x}^2$$

- Add them up:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - 2\bar{x}^2 + \bar{x}^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 - \bar{x}^2$$

$$\mathsf{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2$$

From data to expectations

- For data, "average" means $\frac{1}{N}\sum$: every observation counts $1/N$. For a random variable, "average" means the expectation $\mathbb{E}[\,\cdot\,]$, where every outcome counts with its probability. Same idea.
- Replace each $\frac{1}{N}\sum$ by $\mathbb{E}$ and $\bar{x}$ by $\mathbb{E}[X]$:

$$\frac{1}{N}\sum_{i=1}^{N} x_i^2 - \bar{x}^2 \quad \longrightarrow \quad \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 = \mathsf{Var}(X)$$

- **Check with numbers.** Stock A from Lecture 2 had returns 12, -4, 10, 2, with mean 5. Mean of the squares: $(144 + 16 + 100 + 4)/4 = 66$. Then $66 - 5^2 = 41$. The direct route gave $(49 + 81 + 25 + 9)/4 = 41$. Same number.
- In the lecture we divide by $T - 1$ instead of N for historical returns. The identity still works; only the constant in front changes.

Alternative expressions of the covariance

- **The shortcut formula.** The same trick as König-Huygens works for the covariance: the **mean of the products minus the product of the means**, in expectation and for N observations (x_i, y_i) with means $\bar{x}$, $\bar{y}$:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$$

$$\mathrm{Cov}(X, Y) = \frac{1}{N}\sum_{i=1}^{N} x_i\, y_i - \bar{x}\,\bar{y}$$

- **Covariance from variances.** Take the variance of $X + Y$, remove the two individual variances, and halve what is left:

$$\mathrm{Cov}(X, Y) = \tfrac{1}{2}\left[\mathrm{Var}(X + Y) - \mathrm{Var}(X) - \mathrm{Var}(Y)\right]$$

This is the two-stock portfolio variance formula with weights of 1, read backwards.

Extra 4

The Markowitz problem

Markowitz: the optimization problem

- We have N stocks and must choose the weights $x_1, \ldots, x_N$. Write $\sigma_{ij} = \mathrm{Cov}(R_i, R_j)$ for the covariance between stocks i and j.
- **Target**: the smallest possible portfolio variance.

$$\min_{x_1,\ldots,x_N} \mathrm{Var}(R_p) = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i \, x_j \, \sigma_{ij}$$

- **Constraints**: reach a chosen expected return μ, and invest all the money.

$$\sum_{i=1}^{N} x_i \, E[R_i] = \mu \qquad\qquad \sum_{i=1}^{N} x_i = 1$$

Markowitz: first-order conditions

- Attach a multiplier to each constraint, λ for the return and γ for the budget, and write the Lagrangian (the $\frac{1}{2}$ only cleans up the derivative):

$$\mathcal{L} = \tfrac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} x_i \, x_j \, \sigma_{ij} - \lambda \Big(\sum_{i=1}^{N} x_i \, E[R_i] - \mu \Big) - \gamma \Big(\sum_{i=1}^{N} x_i - 1 \Big)$$

- Partial derivative with respect to each weight x_k, set to zero: one equation per stock.

$$\frac{\partial \mathcal{L}}{\partial x_k} = \sum_{j=1}^{N} \sigma_{kj} \, x_j - \lambda \, E[R_k] - \gamma = 0$$

that is $\quad \mathrm{Cov}(R_k, R_p) = \lambda \, E[R_k] + \gamma$

- The derivatives with respect to λ and γ give back the two constraints: $N + 2$ linear equations for $N + 2$ unknowns.

Markowitz: the solution

- Stack the weights in a column $\mathbf{x}$, the expected returns in a column μ, a column of ones $\mathbf{1}$, and the covariances in the matrix Σ. The N first-order conditions become one line:

$$\Sigma\,\mathbf{x} = \lambda\,\mu + \gamma\,\mathbf{1} \qquad \Longrightarrow \qquad \mathbf{x}^* = \Sigma^{-1}\left(\lambda\,\mu + \gamma\,\mathbf{1}\right)$$

- λ and γ are fixed by the two constraints. With the shorthand $A = \mathbf{1}'\Sigma^{-1}\mathbf{1}$, $B = \mathbf{1}'\Sigma^{-1}\mu$, $C = \mu'\Sigma^{-1}\mu$, $D = AC - B^2$:

$$\lambda = \frac{A\mu - B}{D} \qquad \gamma = \frac{C - B\mu}{D}$$

- In practice: one matrix inverse (`MINVERSE` in Excel) and every optimal portfolio follows.

Markowitz: the frontier

- Plug the solution back into the variance. The smallest variance reachable for a target return μ is

$$\sigma_p^2(\mu) = \frac{A\mu^2 - 2B\mu + C}{D}$$

- A parabola in μ: that is the minimum-variance frontier, and its upper half is the efficient frontier. Its lowest point is the global minimum-variance portfolio, with weights

$$\mathbf{x}_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{A}$$

Markowitz in real life: no closed form

- Asset managers solve the same kind of problem every day, but with extra constraints that a real mandate imposes:
  - **Caps on the weights**: no more than, say, 5% in any single stock, or 25% in any sector.
  - **No short sales**, or a limit on the total short position.
  - **Factor neutrality**: the portfolio's exposure to a risk factor (the market, oil, interest rates) must be zero or within a band.
  - **Turnover and cost limits**: do not trade more than a given fraction of the portfolio.
- With inequality constraints like these there is no formula to plug into. Instead an **optimizer** searches for the weights numerically, and the result is an approximation, accurate to as many decimals as we ask for.

Markowitz in real life: an optimizer

- An open-source example in Python: **CVXPY**. The problem is written almost as it appears on the previous slides: the target to minimise, then the list of constraints, then `solve()`.
- Here with a return target, full investment, no short sales, and a 5% cap on every weight:

```
import cvxpy as cp
x = cp.Variable(N)
constraints = [mu @ x >= target,    # return target
               cp.sum(x) == 1,      # fully invested
               x >= 0,              # no short sales
               x <= 0.05]           # 5% cap per stock
objective = cp.Minimize(cp.quad_form(x, Sigma))
cp.Problem(objective, constraints).solve()
```

- The optimizer returns the weights in `x.value`. Change a constraint, run it again: that is how a real portfolio is rebalanced.